

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5: Measure network performance using key metrics with simulation tools.
(BL4,BL5)

Assessment Tool Weightage: 30%

Annexure A

Pseudocode Writing Task (Algorithm Design)

1. In a network simulation using Cisco Packet Tracer, a student records total data received as 8000 bits in 4 seconds and observes that 200 packets were sent while 180 packets were received. Write a pseudocode algorithm to calculate the network throughput in bits per second and the packet loss percentage based on the given values. The algorithm should include validation to ensure that time and total packets sent are not zero to avoid errors. It should display both the throughput and packet loss results clearly. Add logic to classify the network as Efficient if throughput is greater than 1500 bps and packet loss is less than 10%, otherwise classify it as Inefficient. Finally, ensure the algorithm outputs a complete summary of the network performance based on the calculated metrics.
2. An organization has multiple network links connecting its branches. The administrator wants to monitor bandwidth utilization and identify overloaded links. Write a pseudocode algorithm that periodically collects the bandwidth usage of each network link and stores the values in an array or list. Use a loop to continuously monitor all links. Add conditions to detect when bandwidth utilization exceeds 80% and recommend switching traffic to a backup link. Explain how this algorithm supports efficient bandwidth management and prevents network congestion.
3. A company has six branch offices connected through multiple network links. Each link has different bandwidth, delay, and utilization values, which change throughout the day. The network administrator wants to continuously identify the best communication path between the headquarters and each

branch based on overall link quality rather than just the shortest distance. Write a pseudocode algorithm that periodically collects the bandwidth, delay, and utilization of every network link and stores the values in suitable arrays or lists. Use a loop to repeat the monitoring process at regular intervals. Calculate a link quality score for each path using the collected metrics and select the path with the highest score. Include conditions to detect when a link becomes overloaded or fails, automatically selecting an alternate path and generating an alert for the administrator. Explain how your algorithm improves network reliability, routing efficiency, and fault tolerance while adapting to changing network conditions.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

Q1. Network Throughput and Packet Loss

Answer

Given:

- Total data received = 8000 bits
- Time = 4 seconds
- Packets sent = 200
- Packets received = 180
- Efficient if throughput > 1500 bps AND packet loss < 10%

Pseudocode

BEGIN

INPUT totalDataReceived

INPUT time

INPUT packetsSent

INPUT packetsReceived

IF time <= 0 THEN

 DISPLAY "Error: Time must be greater than zero"

 STOP

END IF

IF packetsSent <= 0 THEN

 DISPLAY "Error: Total packets sent must be greater than zero"

 STOP

END IF

throughput = totalDataReceived / time

lostPackets = packetsSent - packetsReceived

packetLoss = (lostPackets / packetsSent) * 100

DISPLAY "Network Throughput = ", throughput, " bps"

DISPLAY "Packet Loss = ", packetLoss, "%"

IF throughput > 1500 AND packetLoss < 10 THEN

 classification = "Efficient"

ELSE

 classification = "Inefficient"

END IF

DISPLAY "Network Classification = ", classification

DISPLAY "----- Network Performance Summary -----"

DISPLAY "Total Data Received = ", totalDataReceived, " bits"

DISPLAY "Transmission Time = ", time, " seconds"

DISPLAY "Packets Sent = ", packetsSent

DISPLAY "Packets Received = ", packetsReceived

DISPLAY "Lost Packets = ", lostPackets

DISPLAY "Throughput = ", throughput, " bps"

DISPLAY "Packet Loss = ", packetLoss, "%"

DISPLAY "Performance = ", classification

END

Calculation

Throughput:

Lost packets:

Packet loss:

Since the condition requires **throughput > 1500 bps AND packet loss < 10%**, the packet loss condition is not satisfied.

Final Result:

- Throughput = **2000 bps**
- Packet Loss = **10%**
- Classification = **Inefficient**

Q2. Bandwidth Utilization Monitoring

Answer

The algorithm continuously monitors all network links, stores bandwidth utilization values in an array, identifies links exceeding **80% utilization**, and recommends switching traffic to a backup link.

Pseudocode

BEGIN

 INPUT numberOfLinks

 CREATE bandwidthUsage[numberOfLinks]

 IF numberOfLinks <= 0 THEN

 DISPLAY "Error: Number of links must be greater than zero"

 STOP

 END IF

 WHILE TRUE DO

 DISPLAY "----- Bandwidth Monitoring -----"

 FOR i = 0 TO numberOfLinks - 1 DO

 INPUT bandwidthUsage[i]

 IF bandwidthUsage[i] < 0 OR bandwidthUsage[i] > 100 THEN

 DISPLAY "Invalid utilization for Link ", i + 1

 CONTINUE

 END IF

 DISPLAY "Link ", i + 1, " Utilization = ",
 bandwidthUsage[i], "%"

 IF bandwidthUsage[i] > 80 THEN

 DISPLAY "WARNING: Link ", i + 1, " is overloaded"

 DISPLAY "Recommendation: Switch traffic to backup link"

 ELSE

 DISPLAY "Link ", i + 1, " is operating normally"

 END IF

 END FOR

 WAIT for monitoring interval

 END WHILE

END

Working

1. Store the utilization of every link in an array.
2. Use a loop to monitor every link.
3. Check whether utilization is greater than **80%**.
4. If utilization exceeds 80%, generate an overload warning.
5. Recommend switching traffic to a backup link.
6. Continue monitoring at regular intervals.

Example

If the utilization values are:

Link 1 = 55%

Link 2 = 72%

Link 3 = 86%

Link 4 = 61%

The algorithm identifies **Link 3** as overloaded because:

Therefore, traffic should be shifted to its backup link.

This directly satisfies the requirements for periodic collection, arrays, loops, overload detection and backup-link recommendation.

Benefit

The algorithm prevents congestion by detecting overloaded links early and redistributing traffic.

This improves bandwidth utilization and network performance.

Q3. Best Communication Path Selection

Answer

The algorithm monitors **bandwidth, delay and utilization** for all network links. Instead of selecting a path only according to distance, it calculates a **link quality score** and selects the path with the highest score.

Pseudocode

BEGIN

 INPUT numberOfLinks

 INPUT numberOfPaths

 CREATE bandwidth[numberOfLinks]

 CREATE delay[numberOfLinks]

 CREATE utilization[numberOfLinks]

 CREATE linkScore[numberOfLinks]

 CREATE pathScore[numberOfPaths]

 IF numberOfLinks <= 0 OR numberOfPaths <= 0 THEN

 DISPLAY "Error: Invalid number of links or paths"

 STOP

 END IF

 WHILE TRUE DO

 DISPLAY "----- Network Monitoring -----"

 FOR i = 0 TO numberOfLinks - 1 DO

 INPUT bandwidth[i]

 INPUT delay[i]

 INPUT utilization[i]

 IF bandwidth[i] <= 0 THEN

 DISPLAY "Link ", i + 1, " has failed"

 linkScore[i] = 0

 CONTINUE

 END IF

 IF utilization[i] > 80 THEN

 DISPLAY "WARNING: Link ", i + 1, " is overloaded"

 END IF

 linkScore[i] =

 (bandwidth[i] / MAX_BANDWIDTH) * 50

 + ((MAX_DELAY - delay[i]) / MAX_DELAY) * 30

 + ((100 - utilization[i]) / 100) * 20

 END FOR

 FOR p = 0 TO numberOfPaths - 1 DO

 pathScore[p] = 0

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    FOR each link belonging to path p DO
        pathScore[p] =
            pathScore[p] + linkScore[link]
    END FOR

END FOR

bestPath = 0
highestScore = pathScore[0]

FOR p = 1 TO numberOfPaths - 1 DO

    IF pathScore[p] > highestScore THEN
        highestScore = pathScore[p]
        bestPath = p
    END IF

END FOR

IF highestScore = 0 THEN
    DISPLAY "No valid communication path available"
    DISPLAY "Generate emergency network alert"
ELSE
    DISPLAY "Best Communication Path = ", bestPath + 1
    DISPLAY "Path Quality Score = ", highestScore
END IF

FOR i = 0 TO numberOfLinks - 1 DO

    IF bandwidth[i] <= 0 OR utilization[i] > 80 THEN
        DISPLAY "Alert: Link ", i + 1,
            " is unavailable or overloaded"
        DISPLAY "Select alternate path if required"
    END IF

END FOR

WAIT for monitoring interval

END WHILE

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END

Link Quality Formula

A weighted score can be used:

Higher bandwidth increases the score, while lower delay and lower utilization improve the score.

Working

1. Collect bandwidth, delay and utilization.
2. Store values in arrays.
3. Calculate a quality score for each link.
4. Calculate the overall score for each available path.
5. Select the path with the **highest score**.
6. Detect links with utilization above 80%.
7. Detect failed links.
8. Automatically choose an alternate path.
9. Generate an administrator alert.
10. Repeat monitoring periodically.

The algorithm therefore adapts to changing network conditions instead of relying only on shortest distance. This improves **routing efficiency, reliability and fault tolerance**.